SYNTHETIC TEST DOCUMENT

AURORA-WELL-01

A Randomized, Controlled Pilot Study of a Digital Wellness Program for Improving Daily Activity and Sleep Routine Consistency in Adult Volunteers

Figure 1. Conceptual study technology and wellness context. Synthetic visual included to test embedded-image extraction.

Protocol identifier	AURORA-WELL-01
Version	1.1
Date	04 October 2026
Design	Randomized, parallel-group, assessor-blinded pilot study
Population	120 adult volunteers
Sponsor	Northstar Clinical Research Collaborative (fictional)

This protocol is entirely fictional and exists only for software, extraction, chunking, embedding, indexing, image, and table processing tests.

1. Protocol Synopsis

Item	Description
Primary objective	Evaluate whether a structured digital wellness program improves consistency of daily activity over 12 weeks.
Secondary objectives	Evaluate sleep timing regularity, engagement, usability, retention, and data completeness.
Design	Two-week baseline, eight-week intervention, and two-week follow-up; randomized 1:1.
Intervention	Digital reminders, weekly planning prompts, activity goal setting, and sleep-routine education.
Control	Static wellness information without adaptive reminders or weekly planning prompts.
Primary endpoint	Change from baseline in the percentage of evaluable days meeting a participant-specific activity window.
Duration	Up to 14 weeks per participant, including screening and closeout.

1.1 Schedule Overview

Period	Weeks	Main activities
Screening	-2 to 0	Consent, eligibility, demographics, and device compatibility
Baseline	0 to 2	Passive activity capture, sleep diary, and questionnaires
Intervention	2 to 10	Assigned study content, weekly planning, and data capture
Follow-up	10 to 12	Final passive capture, questionnaires, and closeout

The intervention does not provide diagnosis, treatment, emergency monitoring, or medical advice.

2. Background and Rationale

Regular daily activity and consistent sleep timing are common behavioral indicators in wellness research. Routine variability may arise from work schedules, travel, caregiving, and competing demands. Digital planning tools may help participants organize daily routines, although engagement can decline when reminders are excessive or poorly timed.

AURORA-WELL-01 is a low-risk software-enabled pilot study. The program provides general planning prompts rather than a medical intervention. The study is intended to estimate feasibility, retention, engagement, and endpoint variability before a larger investigation.

Figure 2. Conceptual study journey from screening through follow-up and analysis. Synthetic visual included to test image extraction and S3 asset persistence.

2.1 Rationale for the Program

- Weekly planning supports realistic activity and sleep windows.
- Daily reminders are limited and may be snoozed.
- Progress summaries emphasize consistency rather than competition.
- Participants may stop optional prompts at any time.

3. Objectives and Endpoints

Level	Objective	Endpoint
Primary	Assess change in activity regularity	Change in percentage of evaluable days inside the selected activity window
Secondary	Assess sleep timing regularity	Change in standard deviation of nightly sleep midpoint
Secondary	Assess engagement	Proportion of intervention weeks with a completed planning interaction
Secondary	Assess retention	Proportion completing the Week 12 assessment
Secondary	Assess usability	Mean score on the study usability questionnaire
Exploratory	Explore daytime functioning	Change in a short self-reported functioning scale

3.1 Endpoint Rules

- A day is evaluable when at least 10 hours of device availability is recorded.
- Sleep midpoint is derived from participant-entered sleep and wake times after plausibility checks.
- Timestamps are converted to participant-local time.
- Temporary interruptions use available data under the prespecified analysis strategy.

3.2 Sample Size Rationale

A total of 120 randomized participants, approximately 60 per group, is expected to provide adequate precision for feasibility estimates. Assuming up to 15% attrition, approximately 102 participants are expected to contribute to the primary analysis.

4. Study Design and Population

After a two-week baseline period, eligible participants are assigned in a 1:1 ratio to the digital wellness program or control condition. Assignment is stratified by baseline activity regularity and study site.

4.1 Study Flow

Stage	Activity	Output
Screening	Confirm eligibility, consent, and smartphone compatibility	Screened participant
Baseline	Collect at least 7 activity days and 5 sleep diaries	Baseline-qualified participant
Randomization	Apply centralized allocation with concealed variable block sizes	Assigned group
Intervention	Deliver assigned content and collect engagement data	Intervention dataset
Follow-up	Collect final passive data and questionnaires	Completed participant record

4.2 Inclusion Criteria

- Age 21 to 65 years.
- Able to provide informed consent.
- Owns a compatible smartphone.
- Reports an irregular activity or sleep routine.
- Willing to complete brief electronic questionnaires.

4.3 Exclusion Criteria

- Conflicting research participation.
- Planned prolonged travel across more than three time zones.
- Recent use of a similar structured wellness program.
- Investigator determination that participation is unsuitable.

5. Schedule of Activities

Procedure	Screen	Base	Wk 2	Wk 6	Wk 10	Wk 12
Consent	X					
Eligibility	X	X				
Demographics	X					
Device check	X	X				
Activity capture		X	X	X	X	X
Sleep diary		X	X	X	X	X
Randomization			X			
Engagement			X	X	X	
Usability					X	
Functioning scale		X		X		X
Closeout						X

5.1 Data Collection Windows

Assessment	Target	Allowed window
Baseline completion	Day 14	Days 12 to 18
Week 6	Day 42	Days 38 to 46
Week 10	Day 70	Days 66 to 74
Week 12 closeout	Day 84	Days 80 to 91

Missed questionnaires may be reissued within the allowed window. Passive device data are processed using prespecified derivation rules.

6. Data Management, Privacy, and Security

Study data are captured through electronic forms, the study application, and device synchronization services. Automated checks identify missing fields, implausible dates, duplicate submissions, and out-of-window assessments. Corrections require an audit trail.

Figure 3. Conceptual secure data flow across participant devices, cloud services, and research systems. Synthetic visual included to test embedded-image extraction.

6.1 Privacy Controls

- Direct identifiers are stored separately from coded study records.
- Access is restricted by study role and reviewed periodically.
- Data exports use encrypted transfer and controlled storage locations.
- Retention and deletion follow the approved data-management plan.

6.2 Analysis Populations

Population	Definition
Enrolled	All individuals who provide informed consent
Randomized	All participants assigned to a group
Primary analysis	Randomized participants meeting minimum evaluable-data requirements
Per-protocol	Participants without major eligibility or data deviations
Safety summary	Participants who begin the assigned study condition

7. Safety and Quality Oversight

Participants are instructed to report unfavorable experiences associated with study participation. Personnel document reported events, assess seriousness and relatedness, and follow site reporting procedures. The application does not provide emergency monitoring.

7.1 Event Categories

Category	Example	Action
Application issue	Unexpected notification or synchronization failure	Document, troubleshoot, and assess effect on study data
Participation burden	Excessive reminders or questionnaire fatigue	Adjust optional reminders and document preference
Physical experience	Temporary discomfort during voluntary activity	Stop activity and seek appropriate care if required
Privacy concern	Suspected unauthorized access	Escalate under the privacy incident procedure

7.2 Quality Control

- Complete site training before participant-facing activity.
- Use required-field and range checks in key forms.
- Review a sample of consent, eligibility, and endpoint records.
- Resolve or document data queries before database lock.
- Version analysis datasets and derivation specifications.

7.3 Protocol Deviations

A protocol deviation is any departure from the approved procedures or participant instructions. Deviations are documented, assessed for impact on participant rights and data reliability, and reviewed for corrective action.

End of Protocol

This synthetic protocol was generated only to test document upload, native text extraction, image extraction, table extraction, OCR fallback, page offsets, chunking, embedding, and indexing.

Participant Routine Planning Worksheet

Participant ID: _____ **Visit: Week 2**

Complete with study staff after randomization. Keep one copy for reference.

1. Select a realistic daily activity window

Choose a period of the day when ordinary activity is usually possible. The study does not require strenuous exercise. Examples include walking, household activity, or another routine selected by the participant.

- [x] Morning: 06:00 to 10:00
- [ ] Midday: 10:00 to 14:00
- [ ] Afternoon: 14:00 to 18:00
- [ ] Evening: 18:00 to 22:00

2. Select a target sleep routine

Preferred bedtime: _____ Preferred wake time: _____

3. Notification preferences

- [x] Activity reminder enabled
- [x] Sleep reminder enabled
- [x] Weekly planning prompt enabled
- [x] Quiet hours enabled

4. Safety and study reminders

Stop any voluntary activity that causes discomfort. The study application is not an emergency service and does not provide medical advice. Contact an appropriate healthcare professional for health concerns and local emergency services for urgent situations.

Participant initials: _____ Date: _____

Study staff initials: _____ Site: _____

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Protocol Deviation Review Form

Site ID: _____ **Participant ID:** _____ **Date:** _____

Deviation classification

- ☐ Eligibility
- ☑ Visit window
- ☐ Consent
- ☐ Other

Deviation details

Date identified	04 Oct 2026
Procedure affected	Week 6 questionnaire
Expected window	Days 38 to 46
Actual completion	Day 48
Immediate action	Questionnaire accepted and timing deviation documented

Impact assessment

- ☑ No impact on participant rights or safety
- ☑ No material impact on primary endpoint interpretation
- ☐ Corrective or preventive action required

Reviewer conclusion

Minor timing deviation. Retain the assessment in the analysis dataset with the observed completion date. Include the event in the protocol deviation listing and summarize by treatment group before database lock.

Investigator signature

Date

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